

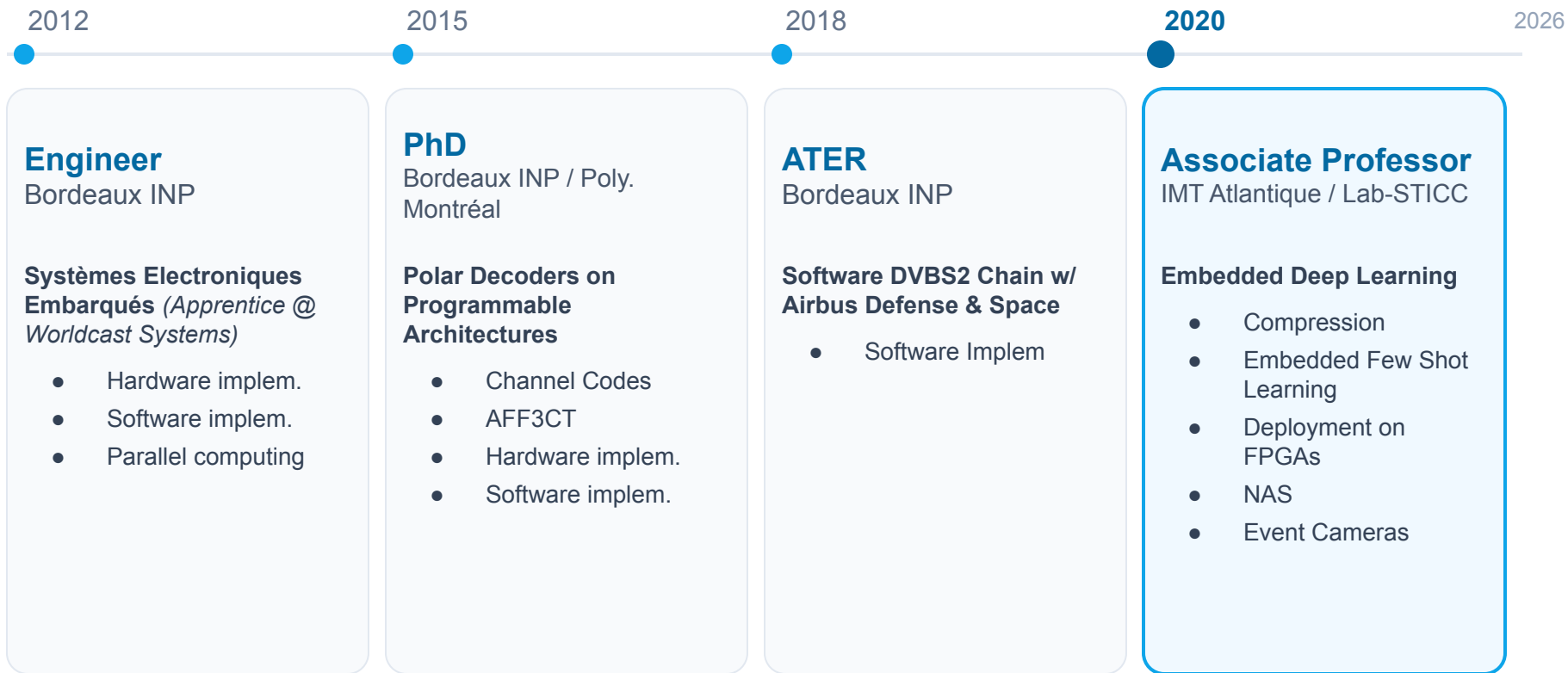
Sobriety through Co-design in Deep Learning and Digital Communications

Habilitation à Diriger des Recherches

Mathieu Léonardon



Timeline



Teachings

~ 170 hours/year face to face (1st, 2nd, 3rd year) • Electronics, HPC, Deep Learning

PARPING

Calcul Parallèle pour l'ingénieur



Innovative teachings



Competition in learning

Prix de la pédagogie IMT 2026

Efficient Deep Learning



Strong link with research



Competition in Learning

BRAIN team

Main research directions

- **Efficient AI:**
embedded systems, compression
- **Steerable AI:**
few shot, zero-shot, adapters
- **Broadened AI:**
complex domains, multi-modality

Main research domains

- Vision (classification/detection...)
- Neuroimaging (decoding/...)
- Audio (classification/detection...)
- Natural Language (foundation...)

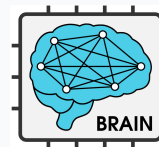
Members

2 Professors, 4 Associate Professors

Axel Marmoret, Bastien Padeloup, Giulia Lioi, Mathieu Leonardon, Nicolas Farrugia, Vincent Gripon

20 PhD Students

Ismail Amessegheer, Yahia Bellouche, Lucas Benoit, Réda Bensaid, Emma Ducos, Carlo Ferritto, Philippe Forest, Arthur Gaudard, Lucie Jean Labadie, Kaiteng Jiang, Gracia Khoury, Karl La Grassa, Elise Lagarde, Ulysse Lefeuvre, Jonathan Lys, Jérémy Morlier, Joshua Riveira, Nolan Sisouphanthong, Venkatesh Subramani, Alexandre Thouvenot



Partners & Institutions

